Practical 2 — Bitwise AND and XOR with 127

1. What is this practical about?

Computers store information using numbers made of 0s and 1s. These are called binary bits.

Your practical studies what happens when we apply two operations, AND and XOR, to characters such as H, e, l, l, o using the number 127.

2. Understand the important words

- Bit: A single binary digit, either 0 or 1.
- Bitwise operation: An operation that works directly on the bits of a number.
- AND (&): The answer is 1 only when both bits are 1.
- XOR (^): The answer is 1 when the two bits are different.
- 127: A decimal number whose 8-bit binary representation is 01111111.

3. Simple example

AND operation

First number: 1100

Second number: 1010

AND result: 1000

Only the positions where both numbers have 1 produce 1.

XOR operation

First number: 1100

Second number: 1010

XOR result: 0110

Different bits produce 1; identical bits produce 0.

4. Why do we use 127?

The binary value of 127 is 01111111.

- Applying AND with 127 keeps the lower seven bits and clears the highest bit in an 8-bit value.
- Applying XOR with 127 flips the lower seven bits.

For example, the character A has ASCII value 65.

A = 01000001

127 = 01111111

AND = 01000001 = 65

XOR = 00111110 = 62

The XOR result, 62, represents the character > in ASCII.

Important: AND and XOR are useful operations in cryptography, but simply applying them to characters does not make a secure encryption system.

Viva answers

Q: What is the aim of this practical?

To study the effect of bitwise AND and XOR operations on characters using the value 127.

Q: What is XOR?

XOR is a bitwise operation that returns 1 when two corresponding bits are different.

Q: Why do we use 127?

Its binary form is 01111111, so it is useful for studying how operations affect the lower seven bits.

Practical 3 — Classical Substitution and Hill Ciphers

1. What is a cipher?

A cipher is a method used to change a readable message into a secret-looking message.

For example:

Original message: HELLO

Encrypted message: KHOOR

Someone who knows the correct key or method can recover the original message.

Your practical covers three methods: Caesar cipher, substitution cipher, and Hill cipher.

2. Caesar cipher

A Caesar cipher replaces each letter with another letter by moving a fixed number of positions in the alphabet.

Imagine the alphabet is a circular track. If you move three positions forward:

Encryption key = 3

HELLO → KHOOR

- H becomes K.
- E becomes H.
- L becomes O.
- L becomes O.
- O becomes R.

To decrypt, move three positions backward.

What is a key? The key is the number of positions by which the letters are shifted. In this example, the key is 3.

Why is Caesar cipher weak? There are only 25 possible non-zero shifts, so an attacker can try them all.

3. Substitution cipher

A substitution cipher replaces each letter with another letter according to a fixed mapping.

For example, imagine the mapping says:

A → Q

B → W

C → E

Then every A becomes Q, every B becomes W, and every C becomes E.

Unlike Caesar cipher, which uses a regular shift, a general substitution cipher can use a more complicated letter mapping.

Main weakness: The same letter always becomes the same replacement letter, which can allow attackers to identify patterns in the message.

4. Hill cipher

The Hill cipher encrypts letters in groups using mathematics and a matrix.

A matrix is a rectangular arrangement of numbers in rows and columns.

Instead of replacing each letter separately, the Hill cipher converts letters into numbers, groups them into pairs, and applies a mathematical formula using a key matrix.

For a two-letter block:

1. Convert the letters into numbers, usually A = 0, B = 1, … Z = 25.
2. Arrange the two numbers as a column.
3. Multiply them by the key matrix.
4. Take each result modulo 26 to keep the values within the alphabet.
5. Convert the resulting numbers back into letters.

To decrypt, the key matrix must have a valid inverse modulo 26.

Viva answers

Q: What is the difference between Caesar and substitution cipher?

Caesar shifts letters by a fixed amount. A general substitution cipher replaces letters according to a chosen mapping.

Q: What is the Hill cipher?

It is a classical cipher that uses matrix multiplication and modulo 26 to encrypt groups of letters.

Q: Why is the Hill cipher key matrix required to be invertible?

Because we need its modular inverse to decrypt the message.

Q: Are these ciphers secure today?

No. They are mainly used to understand the basic concepts of encryption.

Practical 4 — DES Algorithm

1. What is DES?

DES stands for Data Encryption Standard.

DES is an encryption method that protects a message by changing it into unreadable data using a secret key.

Imagine you write a letter and lock it inside a box. Only someone with the correct key can unlock it.

- Original message: Hello World
- Encryption: Changes the message into ciphertext.
- Decryption: Changes the ciphertext back into Hello World.

DES uses the same secret key for encryption and decryption. This is called symmetric encryption.

2. What is a secret key?

A secret key is a value used by the encryption algorithm to protect and recover information.

Think of it like a physical key for a lock. The sender and receiver must have the correct shared key.

3. Important DES facts

Term	**Simple explanation**
Key size	56 effective bits
Block size	64 bits, or 8 bytes
Rounds	16 rounds of processing
Type	Symmetric block cipher
Modern status	Obsolete and insecure

What is a block cipher? It is an algorithm that processes data in fixed-size blocks. DES processes 8 bytes at a time.

What is a round? A round is one stage of repeated mathematical processing used to transform the data.

4. How does your Java program work?

Step 1: Generate a secret key

Step 2: Encrypt Hello World

Step 3: Display the ciphertext in hexadecimal

Step 4: Decrypt with the same key

Output: Hello World

Your program uses these important Java components:

- KeyGenerator: Creates the secret key.
- Cipher: Performs encryption and decryption.
- ENCRYPT_MODE: Tells Java to encrypt.
- DECRYPT_MODE: Tells Java to decrypt.
- doFinal(): Performs the cryptographic operation and returns the result.
- toHex(): Displays the encrypted bytes in hexadecimal form.

Viva answers

Q: Why did we use DES?

To learn the basic working of symmetric encryption and decryption.

Q: Why does the ciphertext change each time?

Because the program generates a new random key for each execution.

Q: What is padding?

Padding adds extra bytes when necessary so that the message fits the cipher's block size.

Q: Which algorithm is preferred over DES today?

AES, which supports stronger keys and is suitable for modern encryption.

Practical 5 — Blowfish Algorithm

1. What is Blowfish?

Blowfish is a symmetric encryption algorithm that uses a secret key to encrypt and decrypt data.

Imagine two people share one key. One uses it to lock a message, and the other uses the same key to unlock it.

Blowfish was designed as a fast alternative to older encryption algorithms.

2. What makes Blowfish different?

- It uses a 64-bit block size.
- It supports variable-length keys.
- It uses the same secret key for encryption and decryption.
- It processes data through multiple rounds of mathematical operations.

A variable-length key means the algorithm can accept keys of different permitted lengths rather than only one fixed length.

3. Example

Suppose you want to send:

Plaintext: Meet me at 5

The program generates a key, encrypts the message, and produces ciphertext. The same key is then used to decrypt the ciphertext.

Message → Blowfish + Secret Key → Ciphertext

Ciphertext + Same Key → Original Message

Your practical uses Java's cryptography library to carry out these operations.

What is ECB mode?

ECB means Electronic Codebook. It encrypts each block independently. It is simple to demonstrate, but it can reveal patterns in repeated data, so it should not be used for protecting real messages.

Viva answers

Q: Is Blowfish symmetric or asymmetric?

Symmetric.

Q: What is its block size?

64 bits.

Q: Why do we generate a key?

The algorithm needs a secret key to encrypt and decrypt the message.

Q: Is Blowfish the best choice for a new application today?

No. Although it is useful for learning, AES-GCM is a more suitable modern choice for authenticated encryption.

Practical 6 — RC4, Blowfish and RSA

This practical compares three different encryption algorithms. The key idea is to understand how they differ.

1. What is RC4?

RC4 is an older encryption algorithm called a stream cipher.

A stream cipher encrypts data one byte at a time, or as a continuous stream, using a generated sequence of bytes called a keystream.

Think of mixing a secret sequence with your message, one character at a time.

RC4 is now considered insecure and should not be used in real applications.

2. What is Blowfish?

As you learned in Practical 5, Blowfish is a symmetric block cipher. It processes data in fixed-size blocks and uses the same secret key to encrypt and decrypt.

3. What is RSA?

RSA is an asymmetric encryption algorithm. It uses two related keys:

- Public key: Can be shared with other people.
- Private key: Must be kept secret.

Imagine you have a special mailbox. Anyone can use the public opening to put a message inside, but only you have the private key to open it.

For the basic RSA encryption demonstration:

Plaintext

↓

Encrypt using public key

↓

Ciphertext

↓

Decrypt using matching private key

↓

Original message

RSA can also be used in digital-signature systems, where the private key signs and the public key verifies.

4. Compare all three

Feature	**RC4**	**Blowfish**	**RSA**
Type	Stream cipher	Block cipher	Public-key algorithm
Keys	Shared secret	Shared secret	Public and private pair
Same key for decryption?	Yes	Yes	No; use matching private key for basic encryption example
Modern security	Insecure	Legacy	Used with suitable secure padding and implementation

Viva answers

Q: What is a stream cipher?

An encryption algorithm that protects data as a stream of bytes using a keystream.

Q: What is a block cipher?

An algorithm that encrypts data in fixed-size blocks.

Q: What is asymmetric encryption?

Encryption based on a public key and a private key.

Q: Why not encrypt a whole large file directly with RSA?

RSA is not designed for efficient bulk-data encryption. A common design uses a symmetric algorithm for the data and public-key cryptography to protect or establish the symmetric key.

Practical 7 — Diffie–Hellman Key Exchange and MITM Attack

1. What is Diffie–Hellman?

Diffie–Hellman is a method that allows two people to create the same secret key over a public communication channel without directly sending that secret key.

Imagine two friends want to agree on a secret colour. They both start with a public colour, mix it with their own secret colour, exchange the mixtures, and combine them with their own secret colour again. Under the scheme's mathematical rules, they arrive at the same final result.

Diffie–Hellman uses mathematics rather than actual colours.

2. How does it work?

Alice

Secret number: a

Bob

Secret number: b

They agree on public mathematical values and exchange public results.

Both calculate a shared secret

Alice uses her private value and Bob's public value; Bob does the reverse.

In the mathematical version, they choose public values p and g.

- Alice chooses a private number a and calculates $A=g^a \bmod p$.
- Bob chooses a private number b and calculates $B=g^b \bmod p$.
- Alice calculates $B^a \bmod p$.
- Bob calculates $A^b \bmod p$.

Both obtain the same shared secret.

What does mod mean? It means the remainder after division. For example, $17 \bmod 5 = 2$.

3. What is a MITM attack?

MITM stands for Man-in-the-Middle attack.

Imagine Alice sends a message to Bob, but an attacker secretly places themselves between them. The attacker may intercept or change messages without either person realizing it.

Unauthenticated Diffie–Hellman can be vulnerable to this attack because it does not, by itself, prove who Alice or Bob actually is.

How can we prevent it? Use an authenticated key exchange, such as one supported by properly validated digital certificates or digital signatures.

Viva answers

Q: Does Diffie–Hellman directly send the secret key?

No. Both parties calculate it independently.

Q: What is a MITM attack?

An attacker secretly intercepts or alters communication between two parties.

Q: What is the purpose of authentication?

To verify the identity of the party we are communicating with.

Q: Is Diffie–Hellman encryption itself?

Its main purpose is key agreement, not directly encrypting the message.

Practical 8 — SSL/TLS Socket and Server

1. What is a socket?

A socket is a communication endpoint that allows two programs to exchange data over a network.

For example, when a chat application sends a message from your phone to a server, the application can use network sockets to communicate.

2. What is a server?

A server is a program or computer that provides a service to another program called a client.

For example:

- Client: Your browser.
- Server: The computer that hosts a website.
- Request: Your browser asks for a page.
- Response: The server sends the page back.

3. What is SSL/TLS?

TLS stands for Transport Layer Security. It protects information travelling between a client and a server.

SSL is the older name associated with this technology. SSL versions are obsolete; modern systems use TLS.

Think of two people speaking through a secure telephone connection. Someone might observe that a call is happening, but the security system is designed to protect the conversation itself.

TLS provides:

- Confidentiality: Outsiders should not be able to read protected data.
- Integrity: Unauthorized changes to the data can be detected.
- Authentication: The server's identity can be checked using certificates.

4. How does your practical work?

1. The server starts listening for connections.
2. The client connects to the server.
3. TLS performs a handshake to agree on security settings and establish keys.
4. The client and server exchange protected messages.
5. The connection is closed.

Important viva answers

Q: What is the difference between a normal socket and a TLS socket?

A normal socket provides network communication. A TLS socket adds cryptographic protection and can authenticate the server.

Q: What is a TLS handshake?

It is the initial process in which the client and server negotiate security settings, establish keys, and authenticate identities as required.

Q: What is a digital certificate?

A digital certificate is a digitally signed document that connects an identity, such as a website name, to a public key.

Q: What is HTTPS?

HTTPS is HTTP communication protected using TLS.

Practical 9 — Telnet and SSL/TLS Packet Analysis

This is the example you mentioned, buddy. Let's understand it from the very beginning.

1. What is Telnet?

Telnet is a network protocol used to communicate with a remote computer through a text-based terminal.

Imagine your computer is in one room and a server is in another city. Telnet lets you send commands to the remote server and receive its responses.

For example, in a controlled lab, a user types a command and the remote machine sends back a response.

The major problem is that traditional Telnet does not encrypt the communication.

If someone captures the traffic on the network, they may be able to read the commands, responses, or credentials being sent.

2. What is SSL/TLS communication?

TLS protects communication by encrypting the application data being transmitted.

For example, when you visit a website using HTTPS, TLS helps protect the data travelling between your browser and the website.

Even if someone captures the packets, the protected application content is normally unreadable without the necessary cryptographic secrets.

3. What is packet analysis?

Packet analysis means examining small units of data travelling over a network.

A packet capture helps us understand which computers are communicating, which protocols they use, and what information is visible.

Imagine a courier transporting parcels:

- Packet capture is like observing the parcels being transported.
- Packet analysis is examining their labels, routes, and contents where accessible.
- Encryption is like putting the contents inside a locked container.

4. What is Wireshark?

Wireshark is a tool that lets us capture and examine network traffic.

It helps us inspect information such as source and destination addresses, protocols, ports, and packet details.

TShark is the command-line version of Wireshark's packet-analysis engine.

5. What are ports?

A port is a number used to identify a particular network service or application endpoint.

For example:

- TCP port 23 is traditionally associated with Telnet.
- TCP port 443 is commonly used for HTTPS over TLS.

These are common defaults, not guarantees that every service uses those ports.

6. What do we do in this practical?

Test 1: Telnet

Capture traffic from an authorized lab connection. Observe that the application data may be readable because it is sent without encryption.

Test 2: TLS

Capture traffic from a TLS connection. Observe the handshake and encrypted application-data records. The actual application content is normally not readable from the capture alone.

The conclusion: Telnet lacks transport encryption, while TLS is designed to protect the confidentiality and integrity of communication.

Viva answers

Q: What is Telnet?

Telnet is a protocol for remote terminal communication that traditionally sends data without encryption.

Q: Why is Telnet insecure?

Because someone who can capture the traffic may read sensitive information transmitted in plaintext.

Q: What is Wireshark?

A network protocol analyser used to capture and inspect packets.

Q: Can Wireshark capture HTTPS packets?

Yes. It can capture TLS traffic, but the protected application data is normally encrypted.

Q: Why do we perform this practical?

To observe the difference between unencrypted Telnet traffic and encrypted TLS communication.

Safety note: Perform packet captures only on your own computer or an authorized laboratory network, using dummy credentials. Never capture someone else's private traffic without permission.

Practical 10 — JSON Web Token (JWT) Locked Web Application

1. What is JWT?

JWT stands for JSON Web Token.

JWT is a compact token format used to carry information between a client and a server. It is commonly used in login systems to help verify that a request comes from a user with a valid token.

Imagine you enter a college library. After verifying your identity, the librarian gives you a temporary entry pass. When you return to a restricted area, the pass is checked before entry is allowed.

A JWT works somewhat like that digital pass, although the server must verify it properly.

2. What is authentication?

Authentication means checking who you are.

For example, you enter your email and password to log in to a website. The server checks whether the credentials are valid.

3. What is authorization?

Authorization means checking what you are allowed to access.

For example, a student can view their own marks, but only an authorized teacher may update them.

4. How does JWT login work?

1. User logs in

Submits username/email and password

2. Server checks credentials

3. Server creates and signs a JWT

4. Client sends the token with later requests

5. Server verifies the token

Valid and authorized → allow access; otherwise → reject

5. What are the three parts of a JWT?

A typical signed JWT contains three parts separated by dots:

header.payload.signature

- Header: Describes the token type and signing algorithm.
- Payload: Contains information called claims, such as a user identifier and expiry time.
- Signature: Helps the server detect tampering and verify that the token was signed using a trusted key.

A very important point: a typical signed JWT is not encrypted. Its payload can generally be decoded by anyone who has the token. Never put passwords or other secrets in it.

6. What happens if a token is invalid?

The server rejects the request instead of granting protected access.

- 401 Unauthorized: The request lacks valid authentication.
- 403 Forbidden: The requester is not allowed to access the resource.

Viva answers

Q: What is JWT?

JWT is a compact token format used to carry claims between parties, often for authentication and access control.

Q: Why do we use JWT?

To let a server verify a token presented with subsequent requests.

Q: What is the signature used for?

To detect tampering and verify the token's authenticity.

Q: Is JWT automatically encrypted?

No. A signed JWT is usually encoded and signed, not encrypted.

Q: Why does a token have an expiry time?

To limit how long the token should remain valid.

Q: What is the difference between authentication and authorization?

Authentication checks who you are; authorization checks what you can access.

Final revision: all your practicals in one table

Practical	What you should remember
2. AND and XOR	Operations on binary bits
3. Classical and Hill ciphers	Shifting letters, replacing letters, and matrix-based encryption
4. DES	Symmetric encryption using a shared secret key
5. Blowfish	A symmetric block cipher
6. RC4, Blowfish and RSA	Stream cipher, block cipher, and public/private keys
7. Diffie–Hellman	Establishing a shared secret; understanding MITM attacks
8. TLS sockets	Protecting communication between a client and server
9. Telnet and TLS analysis	Comparing plaintext traffic with encrypted traffic
10. JWT	Verifying a token before granting protected access

Practical No. 1 is still missing from the photos you sent, so I haven't guessed its topic.

What happens next?

Now that we've established the simple explanation style, the next step is to create a new PDF from these beginner-friendly explanations, rather than the previous dense version. It should include the definitions, everyday examples, step-by-step logic, and viva questions in the same style.

We'll make it something you can actually understand and revise before your exam, buddy! 💪